

Three diagnostics separate a conditional result from a robust claim

A Anchor coverage

all state-years (n=459)



322 interpolated · 1 below · 136 above

2023 states/DC (n=51)



42 interpolated · 0 below · 9 above

within anchors below above

B Boundary-mismatch stress



Added screening allowance (kg CO₂e/MWh)

Blue: CP lower than 1P · Green: 1P lower than CP · two-phase remains first in 51/51.

C Joint assumption-stress ensemble

Air-cooled	0.0%	0.0%	0.3%
Cold plate	0.1%	9.2%	19.1%
One-phase	0.7%	14.6%	25.5%
Two-phase	99.2%	76.2%	55.0%
	narrow	screening	wide

Frequencies describe seeded triangular stress designs, not confidence.

D Functional-unit sensitivity



Adverse correction to two-phase impact
per equivalent useful computation

A correction this small can erase first rank; measured
throughput, server count and lifetime are therefore decision data.